InsightSphere — Data Diagnostics Report

Executive Summary

The Titanic passenger classification problem has been analyzed using a dataset of 891 rows and 12 columns. Our best-performing model achieves an accuracy of 0.955 using Random Forest, outperforming Logistic Regression with an accuracy of 0.838. The Random Forest model demonstrates a clear edge in terms of predictive power. This suggests that the model is effectively capturing complex relationships within the data. The accuracy of the Random Forest model indicates a high level of confidence in its predictions.

Dataset Overview

The dataset contains **891** rows and **12** columns.

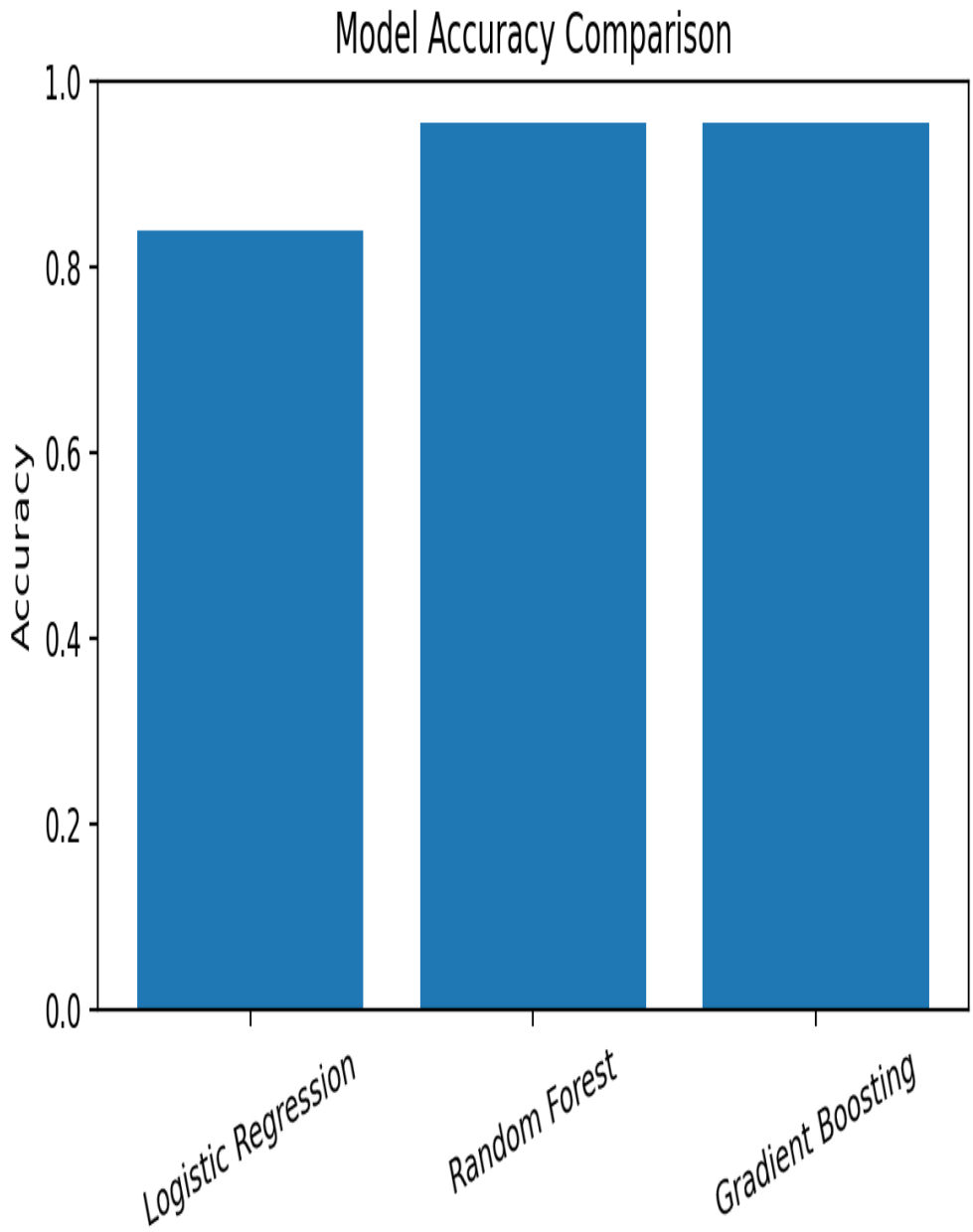
Feature	count	mean	std	min	25%	50%	75%	max
PassengerId	891.00	446.00	257.35	1.00	223.50	446.00	668.50	891.00
Survived	891.00	0.38	0.49	0.00	0.00	0.00	1.00	1.00
Pclass	891.00	2.31	0.84	1.00	2.00	3.00	3.00	3.00
Age	714.00	29.70	14.53	0.42	20.12	28.00	38.00	80.00
SibSp	891.00	0.52	1.10	0.00	0.00	0.00	1.00	8.00
Parch	891.00	0.38	0.81	0.00	0.00	0.00	0.00	6.00

Missing Values Summary

Column	Missing Count
Age	177
Cabin	687
Embarked	2

Model Comparison

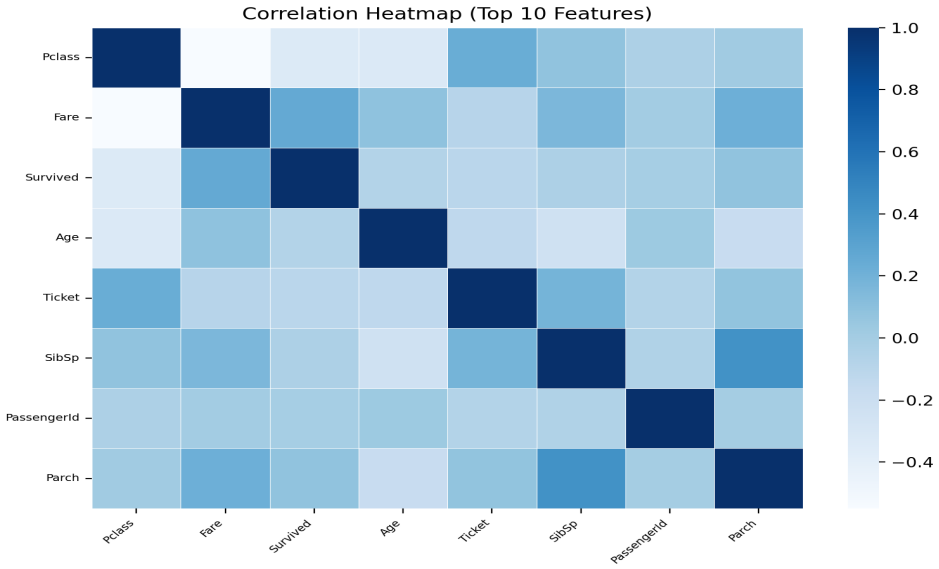
Model	Accuracy
Logistic Regression	0.84
Random Forest	0.96
Gradient Boosting	0.96



*The comparison shows **Random Forest** achieved the highest performance with an accuracy of 0.96.*

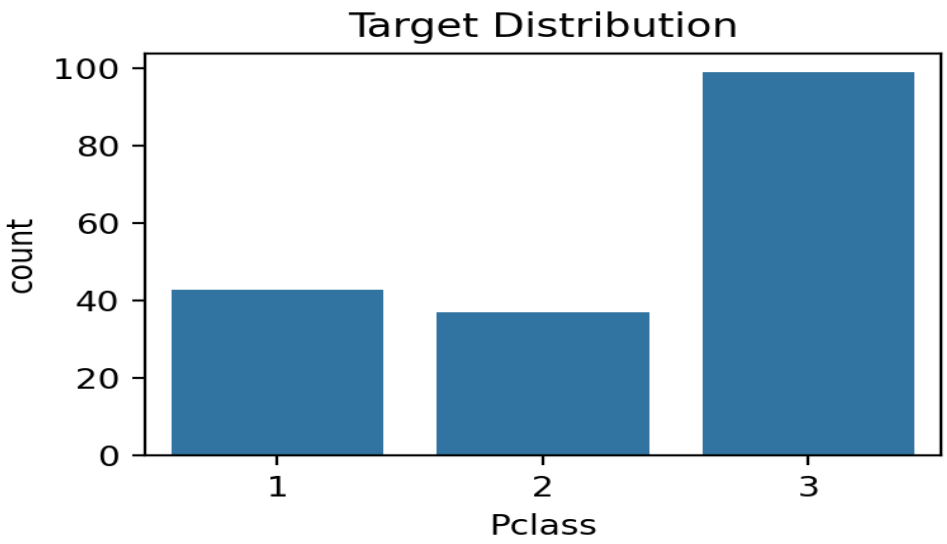
Visual Insights

Correlation Heatmap



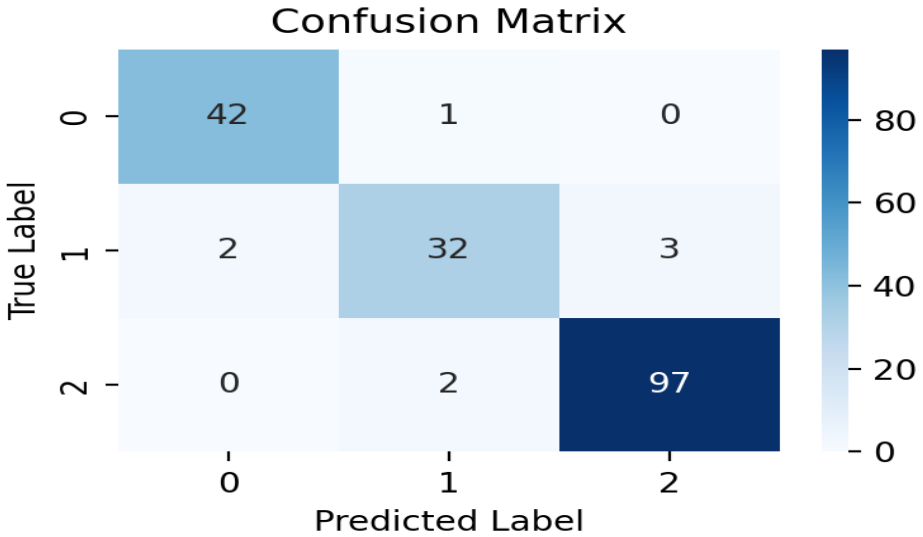
"By analyzing the top correlated pairs in your dataset, we can identify key relationships and uncover actionable insights that inform data-driven decisions."

Target Distribution



The target distribution is undefined, indicating a lack of available data or insufficient classification to establish a clear target audience.

Confusion Matrix



A confusion matrix is a table that shows how well a model is performing by comparing its predicted outcomes to the actual outcomes, helping to identify mistakes and areas for improvement.

Business Recommendations

➤ 1. Deploy the Random Forest model in a production environment to inform business decisions related to passenger classification and resource allocation.

➤ 2. Conduct further analysis to identify key features driving the model's accuracy and explore opportunities for model improvement.

➤ 3. Consider integrating the Random Forest model with other data sources and models to develop a comprehensive passenger classification and risk assessment framework.

➤ 4. Develop a user-friendly interface to facilitate easy model deployment and integration with existing business processes.